Indraprastha WORLD SCHOOL

Paschim Vihar, New Delhi-110063
Session: 2025-26

SAMPLE PAPER (MID TERM EXAMINATION)
MATHEMATICS

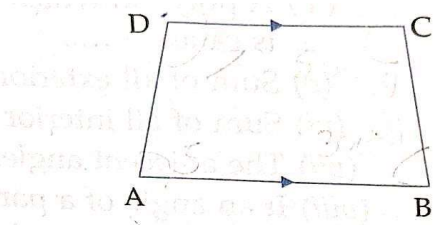
Class: VIII **Maximum Marks: 80**
Date: 00.00.2025 **Time Allowed: 3 hours**

GENERAL INSTRUCTIONS

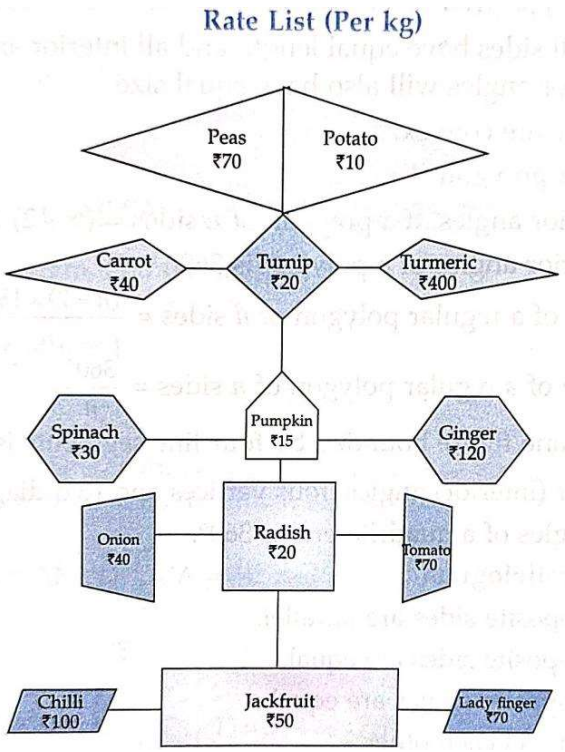
- This question paper contains 38 questions and 5 printed pages.
- This Question Paper is divided into 5 Sections A, B, C, D and E.
- In Section A, Question no. 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion- Reason based questions of 1 mark each.
- In Section B, Question no. 21-25 are very short answer (VSA) type questions, carrying 02 marks each.
- In Section C, Question no. 26-31 are short answer (SA) type questions, carrying 03 marks each.
- In Section D, Question no. 32-35 are long answer (LA) type questions, carrying 05 marks each.
- In Section E, Question no. 36-38 are case study-based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.
- All Questions are compulsory. However, an internal choice in 2 Question of Section B, 2 Questions of Section C and 2 Questions of Section D has been provided. An internal choice has been provided in all the 2 marks questions of Section E.
- Draw neat and clean figures wherever required.
- Use of calculators is not allowed.

	SECTION A **Section A consists of 20 questions of 1 mark each.**	
1.	If a number has 2 in its unit place, then its cube ends in _____. A. 2 B. 1 C. 8 D. 6	1
2.	If a natural number has one zero in its end, then it's cube will have _____ zeros at the end. A. 1 B. 2 C. 3 D. 4	1
3.	Cube root of $1\frac{127}{216}$ is _____. A. $\frac{6}{7}$ B. $-\frac{7}{6}$ C. $1\frac{1}{6}$ D. $-\frac{6}{7}$	1
4.	What is the smallest number by which 75 should be divided to make it a perfect square? A. 1 B. 2 C. 3 D. 4	1
5.	676 is a perfect square. Find the number whose square is 676. A. 46 B. 36 C. 26 D. 16	1
6.	The volume of a cube is $1000\ m^3$. What will be length of its side? A. 30 m B. 20 m C. 10 m D. 21 m	1
7.	Multiplicative Inverse of $\frac{5}{9}$ is _____. A. $\frac{9}{5}$ B. 1 C. $-\frac{9}{5}$ D. $\frac{5}{9}$	1
8.	Square of an odd number is always _____. A. even B. odd C. even or odd D. 0	1
9.	Cost price of a watch is ₹680. If Rishabh sells it at a profit of 35%, then calculate the selling price. A. ₹918 B. ₹919 C. ₹920 D. ₹921	1
10.	What percent is $300g$ of 2 kg? A. 25% B. 35% C. 5% D. 15%	1
11.	What will be the compound interest on ₹10000 at 8% p.a. for 6 months compounded quarterly? A. ₹408 B. ₹10404 C. ₹404 D. ₹400	1

12.	Evaluate $\sqrt[3]{512 \times -125}$ A. 40 B. –40 C. 10 D. –-10	1
13.	Cube root of 13824 is _____. A. 26 B. 28 C. -24 D. 24	1
14.	What is the sum of all interior angles of a 15-sided polygon? A. $2320°$ B. $2340°$ C. $2260°$ D. $360°$	1
15.	Find a Pythagorean triplet whose one member is 12. A. (12,35,37) B. (6,35,37) C. (6,37,38) D. (12,37,38)	1
16.	If ₹12000 taken for 2 years at 4% per annum compounded quarterly, then time period and rate is _____. A. $n = 2, R = 16\%$ B. $n = 4, R = 1\%$ C. $n = 8, R = 1\%$ D. $n = 8, R = 16\%$	1
17.	How many non-square numbers lie between 11^2 and 12^2? A. 24 B. 23 C. 22 D. 21	1
18.	$\sqrt{0.0081}$ is equal to _____. A. 0.09 B. 0.009 C. 0.9 D. 0.0009	1
	DIRECTION: In the question number 19 and 20, a statement of **Assertion (A)** is followed by a statement of **Reason (R)**. Choose the correct option A. Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A). B. Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A). C. Assertion (A) is true but reason (R) is false. D. Assertion (A) is false but reason (R) is true.	1
19.	Assertion (A): 142868 is not a perfect square. Reason(R): A number having 2,3,7 and 8 in its units place cannot be a perfect square.	1
20.	Assertion (A): The cube of 231 is odd. Reason (R): If a number has 1 in its unit's place, then its cube ends in 1.	1
	SECTION-B **Section B consists of 5 questions of 2 marks each.**	
21.	A. The sides of a rectangle are 20 cm and 15 cm. If each side increased by 20%, find the percentage increase in the area. **OR** B. A wardrobe is sold at ₹44080 after allowing a discount of 5%. Find its marked price.	2
22.	Two numbers are in the ratio 4: 5. If the difference of their cubes is 61, find the numbers.	2
23.	A. Verify commutative property of multiplication for $\frac{4}{5}$ and $-\frac{7}{8}$. **OR** B. By what rational number should -3 be divided to get $-\frac{9}{13}$?	2
24.	4225 plants are to be planted in a garden in such a way that each row contains as many plants as the number of rows. Find the number of rows and the number of plants in each row.	2
25.	Each interior angle of a regular polygon is double of its exterior angle. Find the number of sides in the polygon.	2
	SECTION-C **C consists of 6 questions of 3 marks each.**	
26.	Difference of two perfect cubes is 387. If the cube root of the greater of two numbers is 8, find the cube root of the smaller number.	3

27.	A. Find the square root of 9.81 correct to 2 decimal places. **OR** B. Find the greatest four-digit number is a perfect square.	3
28.	One of the diagonals of a rhombus and its sides are equal. Find the angles of the rhombus.	3
29.	Find the cube root of 373248 using prime factorization method.	3
30.	A die is rolled once. Find the probability of getting A. an even number B. a multiple of 3 C. not a multiple of 3.	3
31.	A. A student is trying to verify: $\left(-\frac{3}{2} \div \frac{4}{5}\right) \div -\frac{7}{12} \neq -\frac{3}{2} \div \left(\frac{4}{5} \div -\frac{7}{12}\right)$. Can you simplify both sides and verify if they are not equal? **OR** B. Perimeter of a rectangle is $15\frac{3}{7}$ m. If the length is $4\frac{2}{7}$ m, find its breadth.	3
	SECTION-D **Section D consists of 4 questions of 5 marks each.**	
32.	In the adjoining figure, ABCD is a trapezium. If $\angle A : \angle D = 5:7, \angle B = (3x + 11)°$ and $\angle C = (5x - 31)°$, then find all the angles of the trapezium.  	5
33.	A. A, B, C, D and E went to a restaurant for dinner. A paid $\frac{1}{2}$ of the bill, B paid $\frac{1}{5}$ of the bill and rest of the bill was shared equally by C, D and E. What fraction of the bill was paid by each? **OR** B. If $\frac{3}{5}$ of a number exceeds its $\frac{2}{7}$ by 44, find the number.	5
34.	The following data represents the different number of animals in a zoo. **Animals** | Deer | Tiger | Elephant | Giraffe | Reptiles **Number of animals** | 40 | 10 | 30 | 15 | 25 Prepare a pie chart to represent this data.	5
35.	A. Calculate the amount due and the compound interest on ₹10,000 in 2 years when the rate of interest on successive years is 8% and 9% respectively. **OR** A. Find the compound interest on ₹3125 for 3 years if the rates of interest for the first, second and third year are 4%, 5% and 6% per annum respectively.	5
	SECTION-E **Section E consists of 3 case study based questions of 4 marks each.**	

36. The picture shows a rate list in front of the vegetable shop. The shopkeeper has an educational bent of mind. He wants that children who accompany their parents to purchase vegetables, should get a chance to understand Mathematics also. To every child, he asks one question. To encourage learning, he offers certain discount, in the case, the child answers the question correctly. Yesterday, 4 children were asked questions.



Based on the information given above, answer the following questions:

A. Rashid was asked to name the item written inside the kite shown in the figure. 1

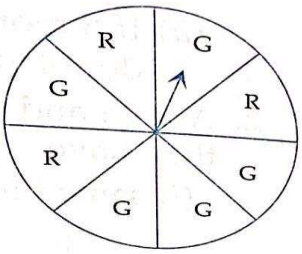
B. Ashu was asked: What is the name of the item written inside the polygon, sum of whose all the interior angles is 720°? Ashu answered it correctly. What was his answer? 1

C. John was asked: What will be the total bill if you buy 2 kg onion, 2 kg pumpkin and 1 kg carrot? 2

OR

C. Niharika was asked: What will be the total price of the items written in all the quadrilaterals shown in the figure above?

37. One day Aarav visited the city sports complex where people were engaged in different activities. There were 200 people in the complex, out of which 40 people were swimming, 35

	people were playing badminton, 25 people were in the gym, and the rest were doing aerobics.  Based on the above information answer the following: A. What percentage of people were swimming? B. What percentage of people was playing badminton? C. What is the fraction of people doing aerobics? **OR** C. What is ratio of rest of the people and people in the gym?	1 1 2
38.	Suppose you spin the wheel shown in adjoining figure.  Based on the picture shown above, answer the following questions: A. What is the total number of possible outcomes of not getting a green sector? B. Find the probability of getting a green sector. C. Find the probability of getting a red sector. **OR** C. Sum of probabilities of getting a green sector and not getting green sector is 1. Verify this statement.	1 1 2